
GENERAL RELATIVITY

Tutorial problem set 11, 29.11.2019.

■ PROBLEM 29 Birkhoff's theorem.

The most general spherically symmetric line element is of the form

$$ds^2 = -A(t, r)dt^2 + 2B(t, r)dtdr + C(t, r)dr^2 + D(t, r)r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2) . \quad (29.1)$$

In problems (a) and (b) you will show that this line element can be reduced to a simpler looking one via coordinate transformation. Therefore, the simpler form of the line element will be just as general as (29.1).

(a) Show that the coordinate transformation of the form

$$r \rightarrow \bar{r} = \bar{r}(r, t) , \quad (29.2)$$

can be chosen such that (after suitable redefinitions) line element reduces to

$$ds^2 = -a(t, r)dt^2 + 2b(t, r)dtdr + c(t, r)dr^2 + r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2) . \quad (29.3)$$

(b) Show that the coordinate transformation of the form

$$t \rightarrow \bar{t} = \bar{t}(t, r) , \quad (29.4)$$

which eliminates the mixed term $dtdr$, always exists and thus the most general spherically symmetric line element (after suitable redefinition of functions) can be written in the form

$$ds^2 = -e^{\nu(t, r)}dt^2 + e^{\mu(t, r)}dr^2 + r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2) . \quad (29.5)$$

Hint: The non-vanishing components of the Riemann tensor for the line element (29.5) are given to you to be,

$$R^0_{101} = \frac{1}{4}e^{\mu-\nu} [2\partial_t^2\mu + (\partial_t\mu)^2 - \partial_t\nu\partial_t\mu] + \frac{1}{4} [\partial_r\nu\partial_r\mu - 2\partial_r^2\nu - (\partial_r\nu)^2] , \quad (29.6)$$

$$R^0_{202} = -\frac{1}{2}re^{-\mu}\partial_r\nu , \quad R^0_{303} = -\frac{1}{2}re^{-\mu}\sin^2\vartheta\partial_r\nu , \quad (29.7)$$

$$R^0_{212} = -\frac{1}{2}re^{-\nu}\partial_t\mu , \quad R^0_{313} = -\frac{1}{2}re^{-\nu}\sin^2\vartheta\partial_t\mu , \quad (29.8)$$

$$R^1_{212} = \frac{1}{2}re^{-\mu}\partial_r\mu , \quad R^1_{313} = \frac{1}{2}re^{-\mu}\sin^2\vartheta\partial_r\mu , \quad R^2_{323} = (1 - e^{-\mu})\sin^2\vartheta , \quad (29.9)$$

and the non-vanishing components of the Ricci tensor are R_{00} , R_{11} , R_{22} , R_{33} , and $R_{01} = R_{10}$. You do not need to calculate the Ricci scalar, why?

After casting the line element in a simpler form (but equally general), in the remaining problems you will determine what functions ν and μ are from the Einstein equation.

- (c) Show that the tr component of the Einstein equation implies that μ is independent of time, $\mu(t, r) = \mu(r)$.
- (d) Use (a linear combination of) the remaining components of the Einstein equation to show that

$$\nu(t, r) = -\mu(r) + f(t) . \quad (29.10)$$

- (e) Use these results to conclude that the most general spherically symmetric solution to the Einstein equation in the vacuum (after a redefinition of the time coordinate) can be written as

$$ds^2 = -e^{-\mu(r)} dt^2 + e^{\mu(r)} dr^2 + r^2 (d\vartheta^2 + \sin^2 \vartheta d\varphi^2) . \quad (29.11)$$

- (f) Show that the other components of the Einstein equation imply

$$e^{-\mu(r)} = 1 - \frac{R_S}{r} , \quad (29.12)$$

where R_S is a real constant, so that we obtain the Schwarzschild solution.

Schwarzschild geometry is the most general asymptotically flat spherically symmetric solution to the Einstein equation. This is the Birkhoff-Jebsen theorem.

■ PROBLEM 30 Newton's potential.

Let us say you have some static spherical distribution of matter (a planet for example). Naturally, one would expect the solution for the metric of such a system to be time-independent, and to possess spherical symmetry (hence use spherical coordinates). Also, far away from the planet, one would expect the space-time to be almost Minkowski.

- (a) Assume the following form for the metric,

$$ds^2 = -[1 + f(r)]dt^2 + [1 + w(r)]dr^2 + r^2 d\vartheta^2 + r^2 \sin^2 \vartheta d\varphi^2 , \quad (30.1)$$

and derive the equations for f and w from the Einstein's equation.

- (b) Show that the leading solution far away from the source reproduces the Newton's potential. Can you find the first correction to Newton's potential?

■ PROBLEM 31 Mitchell's observation.

John Mitchell conjectured in 1783 the existence of objects with black-hole properties. Making use of Newtonian physics, he realized that for every massive object there exists a radius r_S , such that the escape velocity is greater than the speed of light for radial coordinates $r < r_S$. Find the value of this radius as a function of the mass M of the object, the gravitational constant G_N , and the speed of light c .

Do you notice any relation with the radial coordinates of the metric singularities in the Schwarzschild solution?